

GOVT 6019: Homework 3 — Probability

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Part A — The rules of probability

Question 1: Two elections

One registered voter is drawn at random from a county's voter file. Let A be the event that they voted in the spring primary and B the event that they voted in the November general election. In this county,

$$\mathbb{P}(A) = .24, \quad \mathbb{P}(B) = .62, \quad \mathbb{P}(A \cap B) = .21.$$

- State the sample space for this draw, and say what A^c and $A \cap B$ each describe in words.
- Compute $\mathbb{P}(A \cup B)$.
- Compute the probability that the voter cast a ballot in neither election.
- Compute the probability that the voter cast a ballot in **exactly one** of the two. Get to the answer twice: once from your answer to (b), and once by computing $\mathbb{P}(A \cap B^c)$ and $\mathbb{P}(A^c \cap B)$ separately.
- A colleague building the same table reports $\mathbb{P}(A \cap B) = .30$ instead. Explain what the problem with that number is.

Question 2: Regimes and leader exits

Country-years between 1946 and 2020 are cross-classified by regime type and by whether the sitting leader left office through irregular means — a coup, a forced resignation, an assassination. Treat the table as the joint distribution for one country-year drawn at random. (These proportions are stipulated for the exercise.)

	Irregular exit	No irregular exit	Total
Democracy	.002	.548	.550
Autocracy	.018	.432	.450
Total	.020	.980	1.000

- Report the two marginal distributions, and say which cells of the table you summed to get each one.
- Compute $\mathbb{P}(\text{irregular} \mid \text{autocracy})$ and $\mathbb{P}(\text{irregular} \mid \text{democracy})$.

- c. Compute $\mathbb{P}(\text{autocracy} \mid \text{irregular})$.
- d. Your answers to (b) and (c) are both about autocracy and irregular exits, and they are far apart. Explain in two or three sentences what each one measures, and which of the two answers the question “how dangerous is it to be an autocrat?”
- e. Determine whether regime type and irregular exit are independent. Show the comparison the definition requires.

Question 3: Two kinds of sampling

(i) An audit. A county has 40 precincts, 6 of which contain a discrepancy between the machine count and the paper record. An audit selects 3 precincts at random **without replacement**.

- a. Compute the probability that the first precinct selected contains a discrepancy.
- b. Compute the probability that the first two both contain discrepancies.
- c. Compute the probability that **none** of the 3 selected precincts contains a discrepancy, and then the probability that at least one does.
- d. Recompute (c) on the assumption that the three draws are made **with** replacement. Report both numbers to three decimal places, say which procedure is more likely to catch at least one discrepancy, and explain in one sentence why the two answers differ.

(ii) Replications. An early field experiment reports that door-to-door canvassing raises turnout. Five research teams set out to check the result. Each runs its own canvassing experiment in a different city, drawing its own random sample of households and running its own randomization.

For this question, assume the original finding is correct: canvassing does raise turnout. Assume also that each team’s experiment has an 80 percent chance of reproducing it, and that the five experiments are independent of one another. Call a team that reproduces the finding a **success**.

- e. Compute the probability that all five teams succeed.
- f. Compute the probability that at least one of the five does not.
- g. Compute the probability that all k teams succeed for $k = 5, 8, 10, 11, 12$. Then state the smallest k for which that probability falls below .10.
- h. The independence assumed here is a property of how these five experiments were run, not a claim about how politics works. Say where it comes from, then describe one concrete feature a set of replications could have that would break it.

Part B — Bayes’ rule

Question 4: Signature verification

A county processes 200,000 mail ballots. One ballot in 10,000 carries a forged signature. The county’s signature-matching system flags a forged signature with probability .95, and flags a legitimate signature as forged with probability .02.

- a. Let F be the event that a ballot’s signature is forged and Flag the event that the system flags it. Write the quantity the county cares about as a conditional probability, and write out Bayes’ rule for it with the denominator expanded.

- b. Compute $\mathbb{P}(F \mid \text{Flag})$.
- c. Redo the calculation in counts rather than probabilities. Of the 200,000 ballots, how many are forged, how many of those are flagged, how many legitimate ballots are flagged, and how many ballots are flagged in total? Confirm that the ratio of the relevant two counts reproduces your answer to (b).
- d. The county tightens the system so that it flags a legitimate signature as forged with probability .002, leaving everything else unchanged. Recompute $\mathbb{P}(F \mid \text{Flag})$.
- e. Every flagged ballot is set aside until the voter is contacted and confirms it. Using your counts from (c), write two or three sentences on what the base rate implies for that process.

Part C — Random variables and distributions

Question 5: Three coin flips, one experiment

A survey experiment has three subjects. Each subject is assigned to treatment or control by an independent flip of a fair coin. Let X be the number of subjects assigned to treatment.

If a part below has no appropriate answer, say so and explain why.

- a. List the sample space Ω of assignment outcomes, and give the probability of each outcome.
- b. Give the value of X for each outcome in Ω , then write the probability mass function $f(x) = \mathbb{P}(X = x)$ as a table.
- c. Confirm that your answer to (b) satisfies all three conditions in the definition of a PMF.
- d. Write the cumulative distribution function $F(x) = \mathbb{P}(X \leq x)$ at $x = 0, 1, 2, 3$. Then give $F(1.5)$ and $F(-2)$.
- e. Give the probability density function of X .
- f. Name the distribution X follows, with its parameters. (Hint: see slide 44 in the Week 3 lecture slides.)
- g. Compute the probability that the experiment assigns all three subjects to the same condition. Then say, in one sentence, what that number implies about assigning treatment by independent coin flips in an experiment this small.

Part D — Applied: education and party in the VOTER data

```
library(tidyverse)
source("scripts/gov6019_ggplot.R")      # adjust to your project's layout
voter <- read_csv("data/voter2012.csv") # adjust to your project's layout
```

Question 6: A joint distribution estimated from data

Collapse `educ` into a binary variable separating respondents who completed a four-year degree or more from everyone else, then cross-classify by that variable and `pid3`. Drop rows missing either variable, and label both factors with words.

- a. Consult the codebook and say which values of `educ` you put on each side of the cut. How many rows does the complete-case table contain, and how many are dropped? Report how many of the dropped rows are missing `educ`, how many are missing `pid3`, and how many are missing both. State the assumption you make about those respondents by dropping them.
- b. Report the joint distribution as a 2×3 table of relative frequencies over the complete cases, with row and column totals. Confirm that the six cells sum to 1.
- c. Read $\mathbb{P}(\text{Democrat})$ and $\mathbb{P}(\text{BA or higher})$ off your table, and say which cells you summed for each.
- d. Compute $\mathbb{P}(\text{Democrat} \mid \text{BA or higher})$ and $\mathbb{P}(\text{Democrat} \mid \text{No BA})$ from the table. Then compute $\mathbb{P}(\text{BA or higher} \mid \text{Democrat})$.
- e. Assess whether party identification and education are independent in this sample. Compare the six joint relative frequencies against the six products of the corresponding marginals, and report the largest discrepancy. Say which cells are over-represented relative to independence and which are under-represented.
- f. Your answer to (e) is a statement about 7,895 people who answered a survey in 2012. Say what would have to be true for it to be a statement about the American electorate, and name one quantity you would need in order to decide whether a discrepancy of the size you found in (e) is more than an artifact of which 8,000 people happened to be sampled.
- g. Make one figure showing the conditional distribution of party identification given education. Label the factor levels with words, and start any bars at zero. Add a two-sentence caption: the first sentence states what the figure shows, the second states a limitation of the data behind it.

The shares you plot are the ones from (d), not raw counts: the two education groups differ in size, so counts are not comparable across them.

```
# within-group shares: count the cells, then divide by the group's own total
shares <- cc |>
  count(ba_f, pid3_f) |>
  group_by(ba_f) |>
  mutate(share = n / sum(n)) |>
  ungroup()

# side-by-side bars rather than stacked:
#   geom_col(position = position_dodge(width = 0.7), width = 0.6)
# percent labels on the y axis:
#   scale_y_continuous(labels = scales::label_percent(accuracy = 1))
# a title that states the finding, and no axis left unlabeled:
#   labs(title = "...", x = NULL, y = "...")
```

If you want more: Blitzstein and Hwang, *Introduction to Probability* (free at <https://probabilitybook.net/>), chapters 1–3, covers Parts A–C at and somewhat beyond this level; Gailmard, ch. 3, is the assigned treatment; Tversky and Kahneman, “Judgment under Uncertainty: Heuristics and Biases,” *Science* 185(4157), 1974, is the origin of the base-rate problem Question 4 sets up; and the Open

Science Collaboration, “Estimating the Reproducibility of Psychological Science,” *Science* 349(6251), 2015, is the study Question 3(ii) is modeled on.